

HYBRID MODEL FOR PNEUMONIA DETECTION

Parameter-efficient hybrid quantum-classical classifier for chest X-ray pneumonia: simulation and IBM QPU study

Michal Forgó

VISUAL ABSTRACT

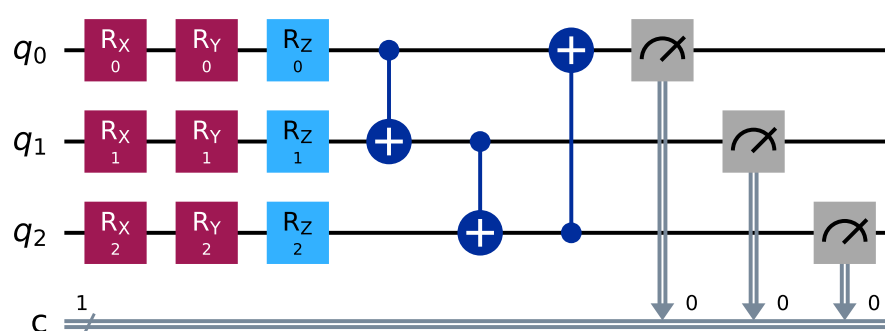
Problem



Limited Expert Availability

Need for Efficient Models

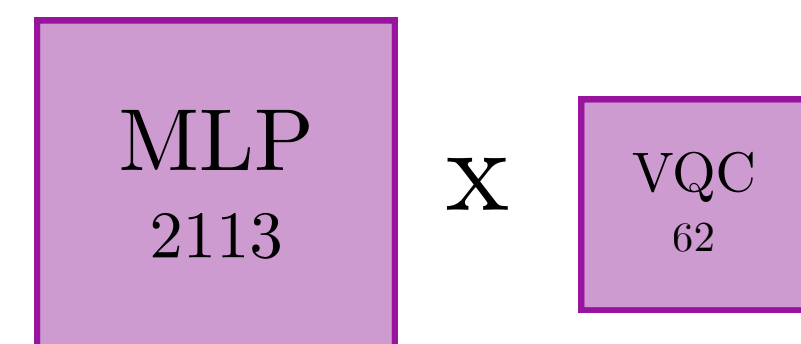
Proposed Solution



Quantum ML Approach

Fair Baseline Benchmarking

Results



34× Fewer Parameters

Distinct Error Patterns

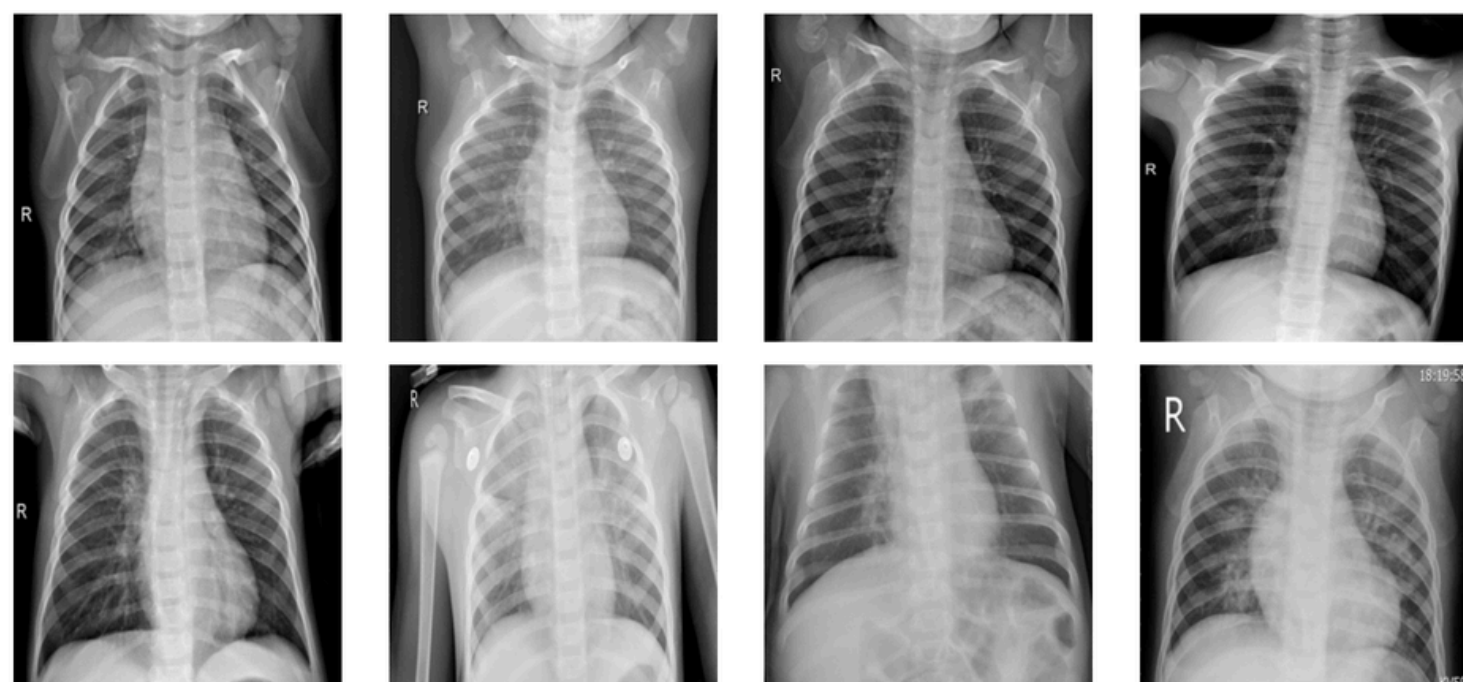
INTRODUCTION

Pneumonia is the leading infectious cause of death among children under five roughly **700,000 lives every year** with the burden concentrated in low-resource settings. Clinical signs are not specific, so diagnosis depends on chest X-ray (CXR) interpretation: a task that is operator-dependent, time-consuming, and constrained by a **worldwide shortage of expert radiologists**.

Machine-learning classifiers can assist CXR triage, but accurate deep models carry a steep cost millions of trainable parameters, **black-box decisions**, and **high computational demand** and they degrade under realistic clinical conditions: dataset shift, class imbalance, and patient overlap between training and test sets.

This project explores compact **hybrid quantum-classical** classifiers: classical deep feature extraction paired with a small trainable quantum circuit as a route toward pulmonary diagnostics that are **accurate, parameter-efficient, and more**.

Data Showcase



RESEARCH OBJECTIVES

O1: Hybrid Pipeline

Integrate ConvNeXt-Tiny and S-VAE extraction with a 6-qubit VQC.

O2: NISQ Optimization

Apply L=3 re-uploading and ZNE to maximize hardware efficiency.

O3: Realistic Bench

Evaluate generalization using patient-level splits and intentional domain shifts.

O4: Trade-off Analysis

Prove statistical parity against a classical MLP using McNemar's test.

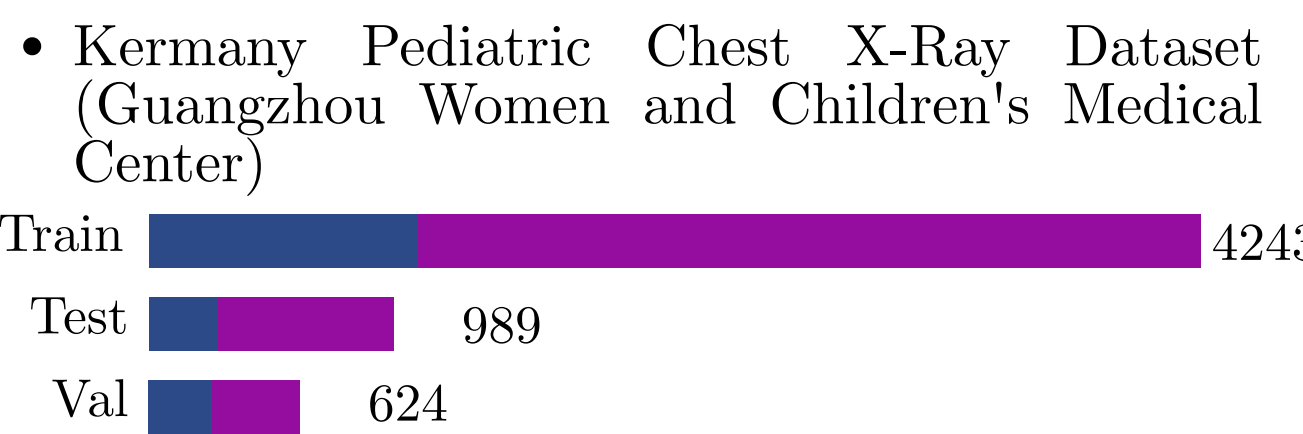
Research Hypothesis

A 6-qubit Variational Quantum Classifier can achieve statistically indistinguishable diagnostic parity with a dense classical Multilayer Perceptron on complex clinical imaging.

METHODOLOGY

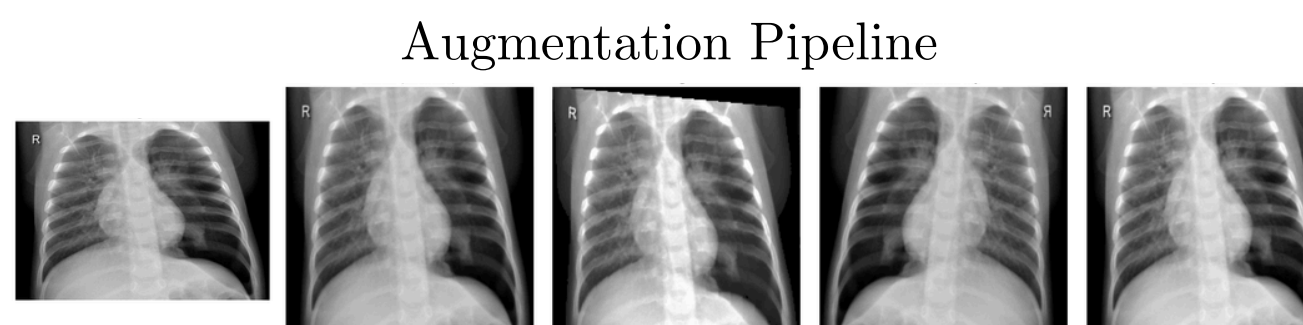
Clinical Input Data

Raw Radiography Imagery (224 x 224)



Clinical Preprocessing

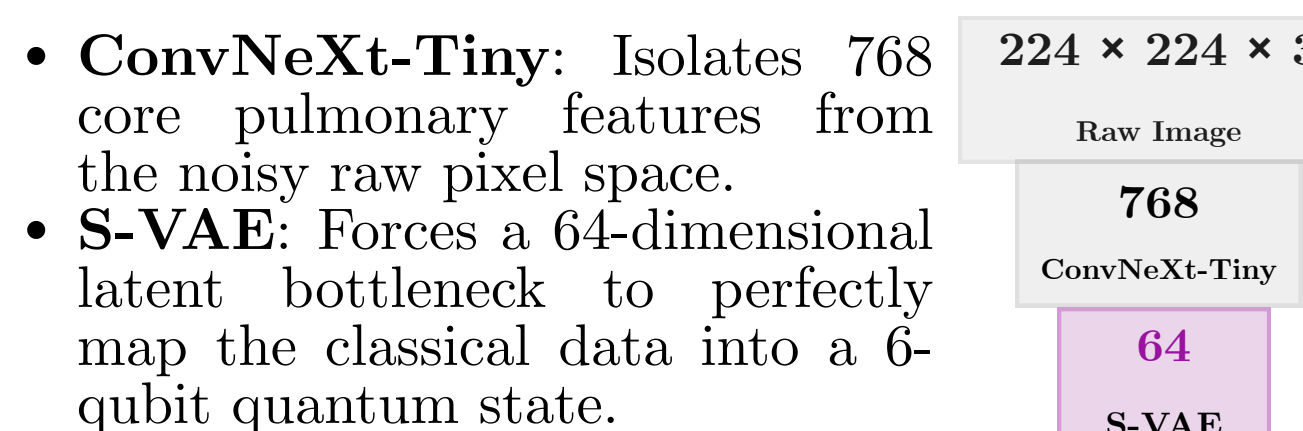
ImageNet Norm. & Med-Safe Augmentation



- Standardization:** Tensors resized to 224 x 224 with ImageNet channel normalization to align with ConvNeXt pre-training.
- Med-Safe Transforms:** Pipeline limited to mild affine transformations and color jitter to expand the training distribution safely.

Feature Extractor

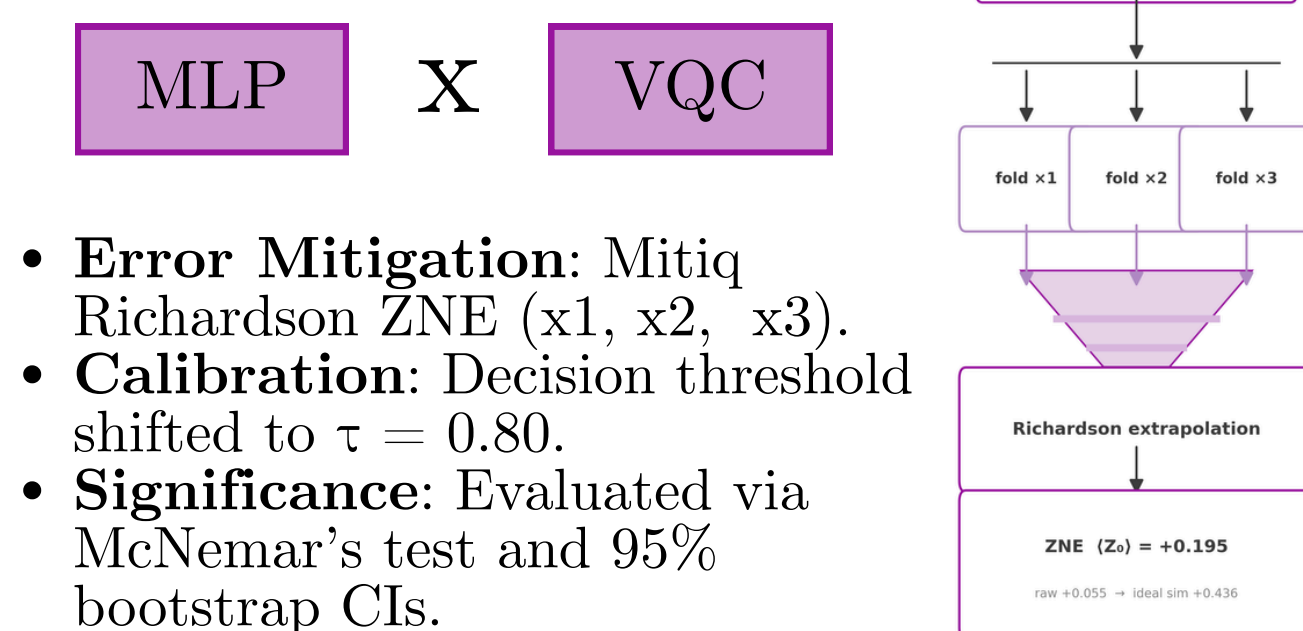
768-dim to 64-dim Tensor Compression



- ConvNeXt-Tiny:** Isolates 768 core pulmonary features from the noisy raw pixel space.
- S-VAE:** Forces a 64-dimensional latent bottleneck to perfectly map the classical data into a 6-qubit quantum state.

Classifier Benchmarking Setup

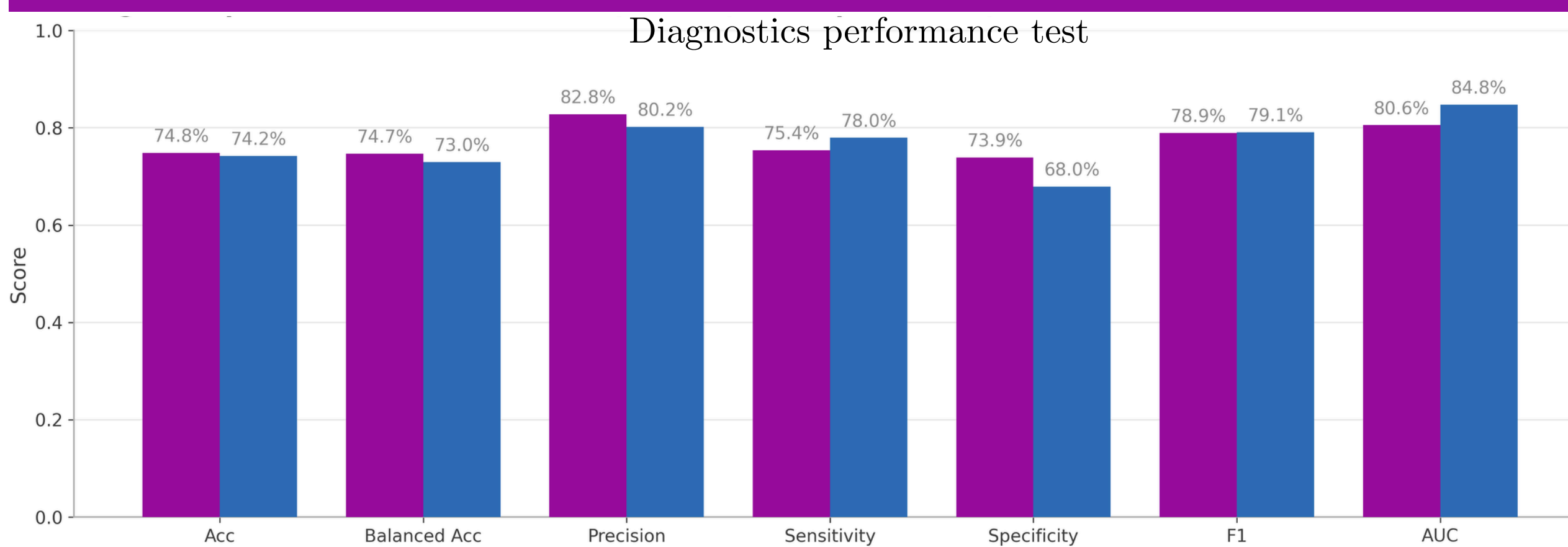
Post-Processing & Evaluation



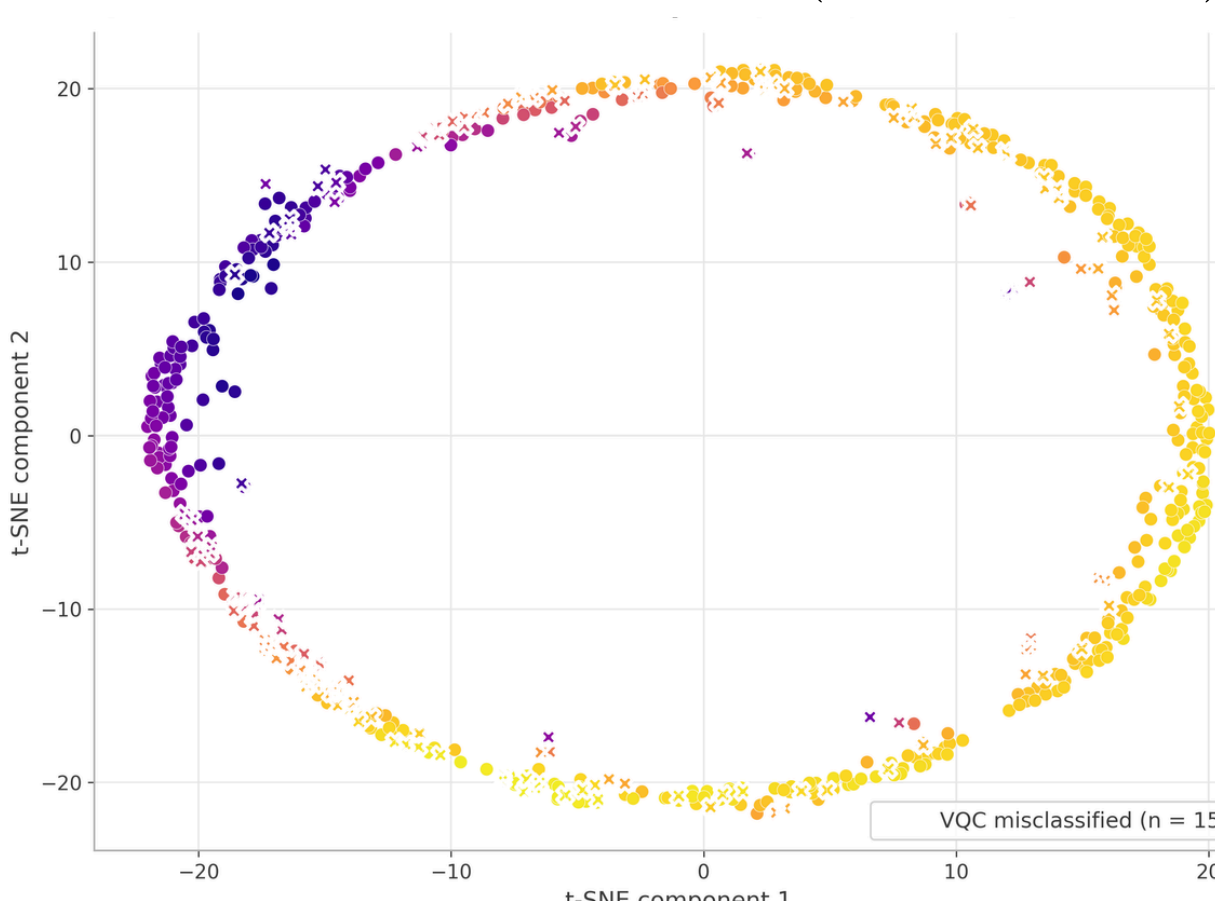
- Classical MLP:** Dense baseline requiring 2,113 parameters.
- Quantum VQC:** Reaches accuracy parity with just 62 parameters via L=3 data re-uploading.
- Error Mitigation:** Mitiq Richardson ZNE (x1, x2, x3).
- Calibration:** Decision threshold shifted to $\tau = 0.80$.
- Significance:** Evaluated via McNemar's test and 95% bootstrap CIs.

RESULTS

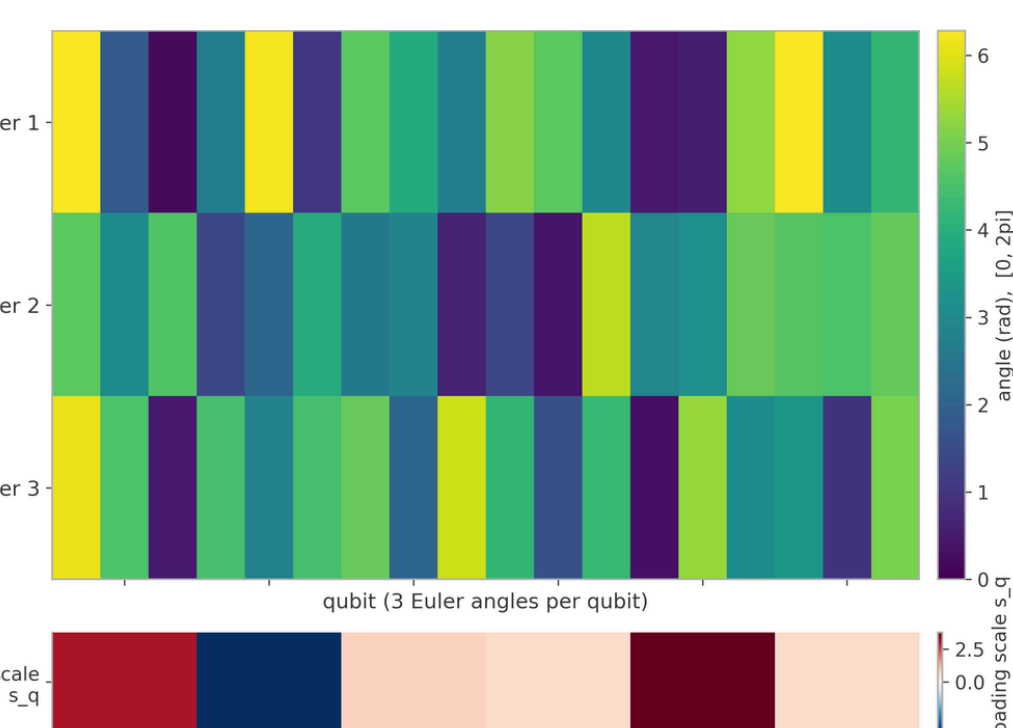
Diagnostics performance test



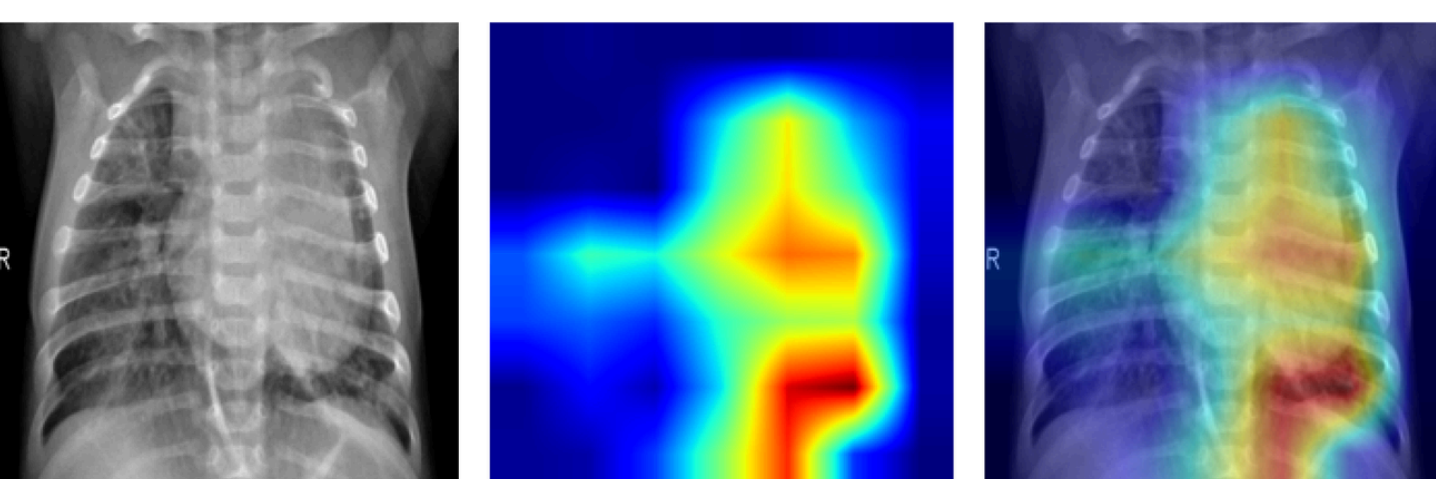
Quantum decision field (t-SNE, seed=6)



Trained rotation angles

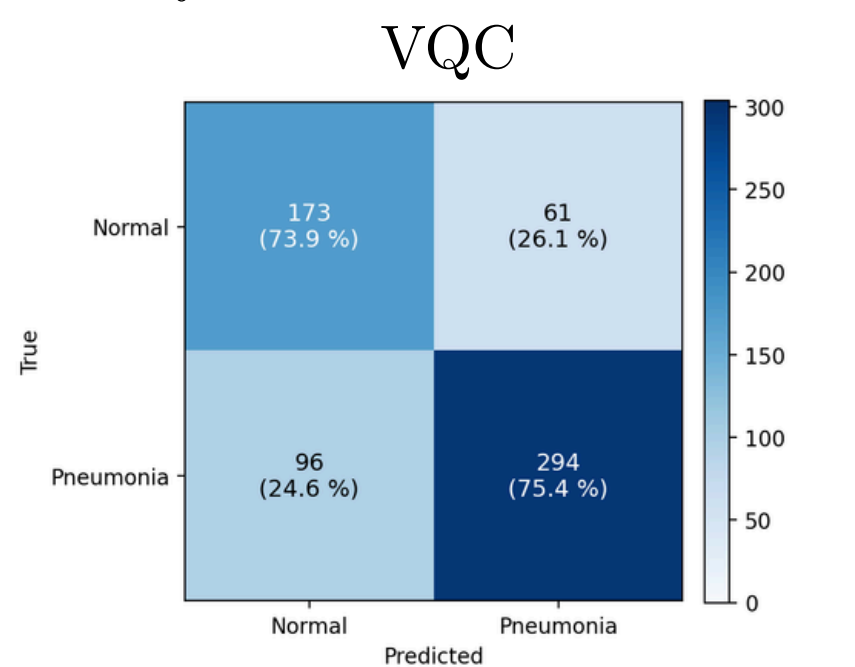


- Latent Topology:** The S-VAE successfully compresses the 768-dimensional ConvNeXt features into a 64-dimensional latent space with clear class separation.
- Quantum Decision Boundary:** The t-SNE projection confirms the 6-qubit VQC successfully maps the non-linear topology of the classical data, achieving a classification threshold ($\tau=0.80$) that mirrors the MLP.



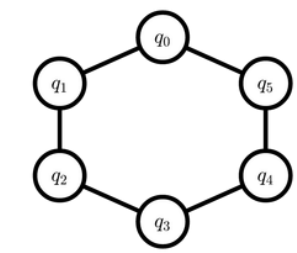
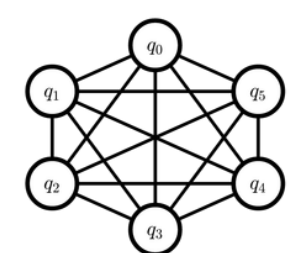
Grad-Cam Showcase

- Parameter Distribution:** Trained Euler angles across three layers show broad, active optimization.
- Dynamic Scaling:** Six distinct re-uploading parameters actively weight the classical feature inputs.
- Amplitude Encoding:** The 64-dimensional features successfully map into distinct 6-qubit probability states.



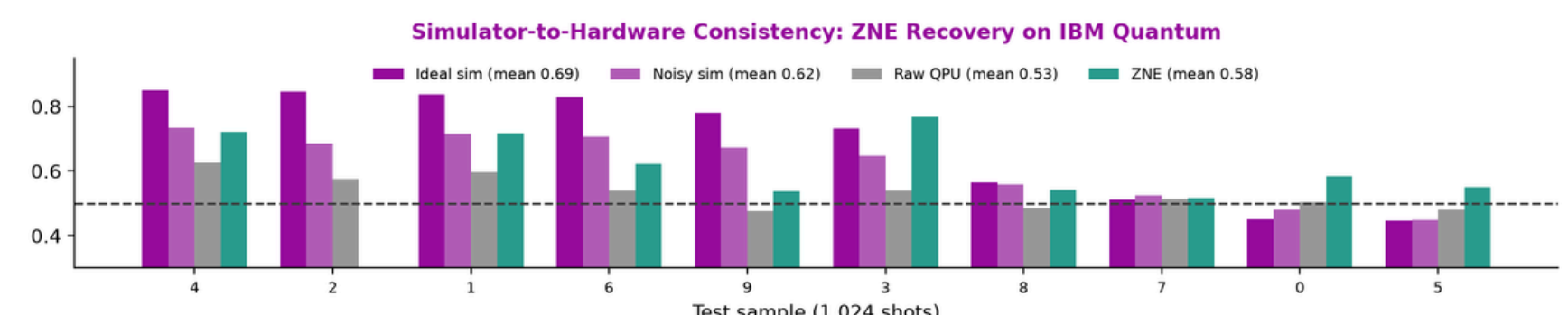
QPU vs SIMULATOR

The ansatz entangles every qubit with its nearest neighbour in a **closed ring** one CNOT per qubit, just **6 two-qubit gates per layer**. This is a deliberate NISQ-era choice: full **all-to-all** connectivity would need **15 CNOTs per layer** (2.5× more), and a ring mirrors the coupling maps of real IBM processors, so the circuit transpiles with minimal **SWAP overhead**. Despite the sparse topology, measured entanglement capability reaches $\text{Ent}(\mathbf{A}) = 0.95$ near the theoretical maximum. The ring proves that an **informationally rich** circuit does not need to be dense, and shallow circuits are exactly what survive on noisy hardware, since **decoherence** grows with depth.



Rather than reading out the **full six-qubit state**, the classifier measures **one expectation value**, $\langle Z_0 \rangle$, on a single qubit preceded by a trainable basis rotation ($\text{RY} + \text{RZ}$, 2 parameters). This concentrates the entire 1,024-shot budget on one decision variable, **minimizing readout error** (the dominant noise source on IBM processors) and maximizing shot efficiency. The learnable basis lets the circuit adapt its measurement axis to the data instead of committing to a fixed Z-basis readout the quantum analogue of learning the optimal linear projection before a decision, at a cost of just 2 of the model's 62 parameters.

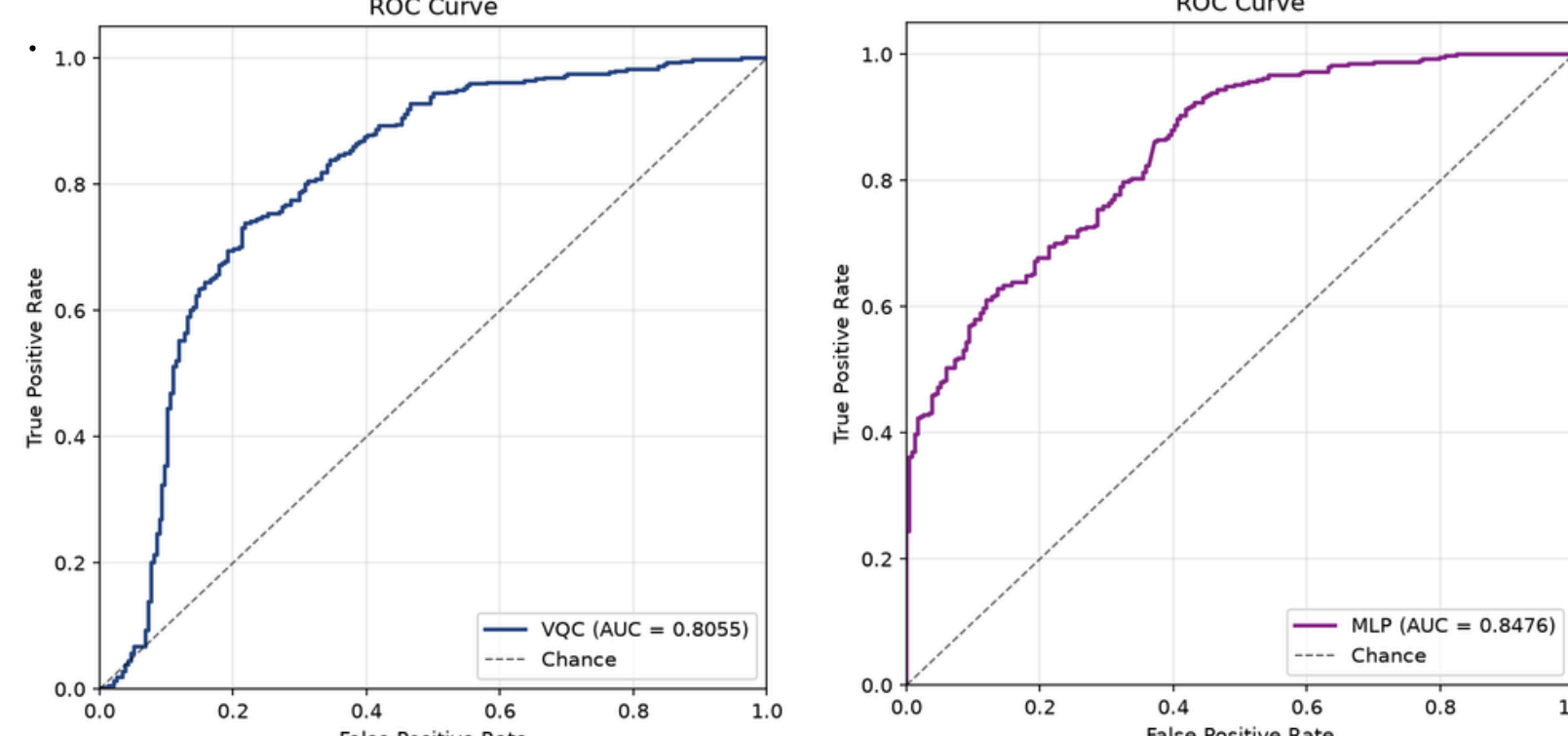
Zero-Noise Extrapolation (ZNE) runs the circuit at **amplified noise** ($\times 1, \times 2, \times 3$ gate folding) and extrapolates to the zero-noise limit no extra qubits. On our IBM Quantum run, Richardson extrapolation (Mitiq) lifted the decision variable toward simulation ($0.53 \rightarrow 0.58$) and cut deviation from ideal ($0.169 \rightarrow 0.163$) signal partially recovered **at zero hardware cost**.



CONCLUSIONS

The central **hypothesis was supported**: a 6-qubit VQC reaches diagnostic performance no statistically significant difference detected (McNemar $p = 0.71$, $n = 624$) from a classical MLP at **34× fewer trainable parameters** (62 vs. 2,113).

- Comparable accuracy:** balanced accuracy 74.7 % (VQC) vs. 73.0 % (MLP); McNemar's test not significant ($p = 0.71$) and bootstrap CIs overlap.
- Honest trade-off:** the MLP retains a higher AUC (0.848 vs. 0.806) the quantum layer trades a small margin of discrimination for extreme parameter efficiency.
- The NISQ bottleneck is real:** a run on IBM Quantum showed noise collapsing the signal by $\sim 52\%$ — current hardware, not the algorithm, limits deployment.



Overall: hybrid quantum-classical learning is viable for medical imaging in the NISQ era parameter-efficient, statistically sound, and worth pursuing as hardware matures.

FUTURE WORK

The study opens three clear directions, each addressing a limitation of the current NISQ-era prototype:

- Hardware.** Scale Zero-Noise Extrapolation (ZNE) and Probabilistic Error Cancellation (PEC) on IBM Quantum closing the simulation-to-device gap measured in our noise experiments.
- Generality.** Validate on multi-institutional datasets (NIH ChestX-ray14, CheXpert) to test robustness under real hospital-to-hospital domain shift.
- Architecture.** Explore quantum kernels, quanyvolutional layers, and classical-quantum ensembles asking whether the 34× parameter advantage can turn parity into a diagnostic edge.

